

**GC University Lahore**

**Examination: Final Semester Summer 2017**

**Subject: Database Systems**

**Course Code: CS-2014**

**Allowed Time: 155 Minutes**

**Max Marks: 46**

**SECTION: OBJECTIVE TYPE**

**Max Marks: 12**

**Instruction: ENCIRCLE the most appropriate option for the following statements.**

- 1. To eliminate duplicate records \_\_\_\_\_ is used.**
  - ☐ A. Eliminate
  - ☐ B. Remove
  - ☐ C. Unique
  - ☐ D. Distinct
- 2. \_\_\_\_\_ SQL command can be used to modify existing data in a database table.**
  - ☐ A. Modify
  - ☐ B. Alter
  - ☐ C. DRL
  - ☐ D. Update
- 3. Which one of the following statements is part of DDL commands?**
  - ☐ A. Insert
  - ☐ B. Drop
  - ☐ C. Create
  - ☐ D. Both B and C
- 4. The Number of entity types that participate in a relationship is known as \_\_\_\_\_**
  - ☐ A. Cardinality
  - ☐ B. Relations

- C. Degree
  - D. None of above
5. For each attribute of relation R, there is a set of allowable values, called the \_\_\_\_\_ of that attribute.
- A. Relationship
  - B. DML
  - C. Domain
  - D. Query
6. If an instance of relational schema  $R(A,B,C)$  have distinct values of  $A$  excluding NULL values, then \_\_\_\_\_
- A. A is a secondary key
  - B. A is non-key
  - C. A is primary key
  - D. A is foreign key
7. An attribute is atomic/simple if elements of the domain are considered to be \_\_\_\_\_ Units
- A. Different
  - B. Divisible
  - C. Indivisible
  - D. Both A and C
8. One can use Max and Min functions for \_\_\_\_\_ data types.
- A. Numeric
  - B. Character
  - C. Date
  - D. All of above
9. If every non-key attribute is fully functional dependent on the primary key, the relation will be in \_\_\_\_\_ normal form
- A.  $1^{\text{st}}$
  - B.  $2^{\text{nd}}$

- C. 3rd
  - D. BCNF
10. \_\_\_\_\_ database level is closest to the users.
- A. Internal
  - B. User level
  - C. External
  - D. Conceptual
11. \_\_\_\_\_ Least time grain of DWH is feasible for organizations working 24/7.
- A. Monthly
  - B. Weekly
  - C. Daily
  - D. Hourly
12. Do not display non matched tuples \_\_\_\_\_
- A. Left Outer Join
  - B. Full Outer Join
  - C. Cross Join
  - D. Inner Join

#### SECTION: SUBJECTIVE TYPE

Max Marks: 46

Attempt any 4 questions.

Question # 1: Marks (12)

A College consists of 52 university and departmental social societies. The societies have students from different departments. The student record contains the student name, reg, phone no & address that includes house no, street no, city, society designations, and department. Each department has a name, location, and dept\_id. Every society has exactly one logo to access the society events. There are different events managed by the society that is sponsored by different sponsors. The event information contains the event id, event name and event sponsor details. But it is not necessary that all the events may have more than one sponsor. The sponsor information contains the sponsor Name, Id, and Address. Finance section is managed by Finance secretary of each society who will finance the contains the

finance id, amount and the event on which finance is consumed. One society can manage many events whereas an event exclusively belongs to a society. One student belongs to only one department whereas a department has many students studying in it. As per societies are concerned, any student can become member of as many societies as he wishes. The society has its id, name, level, and estd. date.

**Note: State clearly any other VALID assumptions that you make.**

- a. Perform rough venn analysis and draw ERD. (4+5)
- b. Make relational model/ form ERD developed in part a. (3)

**Question # 2: Marks (12)**

- a. Define Primary key, Foreign key, Composite key and candidate key. (4)
- b. Define relationship. Differentiate between specialization and generalization. (4)
- c. Define entity. Explain unary M:N relationship by the help of example. (4)

**Question # 4: Marks (12)**

- a. Define and elaborate deadlock. How concurrency can be managed using 2 Phase locking scheme? (2+4)
- b. Define anomalies and explain different types along with suitable example. (6)

**Question # 5: Marks (12)**

- a. Define Data warehouse, How data warehouse model addresses the issues of dependent and independent data architecture? (1+5)
- b. What are properties of transaction? Elaborate the "inconsistent retrieval" issue in the absence of concurrency control. (2+4)

**Question # 6: Marks (12)**

NBN is having trouble in managing their record. You are hired as analyst to give solution for the issue they are facing by normalizing the data up-to 4NF (1NF, 2NF, 3NF, BCNF, 4NF) and resolving relationships. Elaborate each step you performed.

| M_id | Movie Title | Year | Language                     | C_id | Name    | Phone | CNIC  | R_id | D_date | R_date | P_id | P_name      | Country |
|------|-------------|------|------------------------------|------|---------|-------|-------|------|--------|--------|------|-------------|---------|
| M001 | Silence     | 120  | English, Urdu, Dard, Punjabi | C001 | Nadeem  | 0321  | 352-1 | R001 | 6-6-12 | 6-6-17 | P001 | Fly Films   | USA     |
|      |             |      |                              | C004 | Naimat  | 0323  | 352-5 | R002 | 6-7-12 | 4-7-17 |      |             |         |
|      |             |      |                              | C002 | Numai z | 0302  | 352-2 | R003 | 7-6-12 | 7-6-12 | P002 | Linger gate | USA     |
| M002 | Arrival     | 90   | English                      | C001 | Nadeem  | 0321  | 352-1 | R004 | 5-6-12 | 5-6-17 | P003 | CBS Films   | USA     |

| M_id | Movie Title | Year | Language         | C_id | Name   | Phone | CNIC  | R_id | D_date | R_date | P_id | P_name   | Country |
|------|-------------|------|------------------|------|--------|-------|-------|------|--------|--------|------|----------|---------|
| M003 | Passengers  | 120  | English, Urdu    | C003 | Naheem | 0303  | 352-3 | R005 | 5-7-12 | 4-7-17 | P004 | T-series | UK      |
| M004 | Sully       | 96   | English, Chinese | C001 | Nadeem | 0321  | 352-1 | R006 | 7-6-12 | 7-6-12 |      |          |         |

#### Keys Description:

- **M\_id:** Movie ID
- **Lang:** Language
- **C\_id:** Customer ID
- **R\_id:** Date = Returned Date
- **P\_id:** Producer ID
- **Country:** Producer Country
- **CNIC:** Computerized National ID Card
- **R\_id:** Rental ID
- **P\_Name:** Producer Name